

Circular Economy Ontology Network (CEON) - Resource ODP

Metadata

IRI

<http://w3id.org/CEON/ontology/resourceODP/>

Title

Circular Economy Ontology Network (CEON) - Resource ODP

Creator

Huanyu Li

Contributor

Eva Blomqvist

Mikael Lindecrantz

Robin Keskisärkkä

Date Created

2023-03-16

License

<https://creativecommons.org/licenses/by/4.0/>

Version Iri

<http://w3id.org/CEON/ontology/resourceODP/0.2/>

Version Info

0.2

Preferred Namespace Prefix

resourceODP

Preferred Namespace Uri

<http://w3id.org/CEON/ontology/resourceODP/>

Description

A core ODP of the CEON ontology network defining aspects of the resource concept.

Covers Requirements

Covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-1,3, CVN-Composition-1,2, CVN-ResrouceType-4, C7-3, E2-2, E4-6, E5-1, E6-3, T3-1.

Classes

Batch of objects ^c	
IRI	http://w3id.org/CEON/ontology/resourceODP/BatchOf0bjects
Description	A batch of objects is a collection of physical objects that are of the same type, e.g. a set of items (product objects) adhering to the same product model.
Sub Class Of	Resource ^c has physical object ^{op} only Physical object ^c and has physical object ^{op} some Physical object ^c
In Domain Of	batch size ^{dp}
In Range Of	has batch ^{op}
Restriction	batch size ^{dp} <i>exactly</i> 1 Batch of objects ^c

Composition^c

IRI	http://w3id.org/CE0N/ontology/resourceODP/Composition
Description	A composition is used to represent how a whole or mixture is made up. E.g., for chemical elements composing a chemical structure, the composition should hold information of the ratio of the composing chemical elements.

Constituent^c

IRI	http://w3id.org/CE0N/ontology/resourceODP/Constituent
Description	A constituent is a component of object.
In Range Of	has constituent ^{op}

Digital object^c

IRI	http://w3id.org/CE0N/ontology/resourceODP/DigitalObject
Sub Class Of	Resource ^c

Information^c

IRI	http://w3id.org/CE0N/ontology/resourceODP/Information
Description	Information is an abstract concept that represents any kind of interpretations. For instance, information can be data generated by software systems or data used by people for communications.
Sub Class Of	Resource ^c
In Domain Of	containsInformation ^{op} isAbout ^{op}
In Range Of	containsInformation ^{op} is realization of ^{op}

Matter^c

IRI	http://w3id.org/CE0N/ontology/resourceODP/Matter
Description	A matter is a physical substance.
In Range Of	has matter ^{op}

Physical object^C

IRI	http://w3id.org/CE0N/ontology/resource0DP/PhysicalObject
Description	A physical object is a collection of matter.
Sub Class Of	Resource^C has constituent^{op} only Constituent^C <i>and</i> has constituent^{op} some Constituent^C has matter^{op} only Matter^C <i>and</i> has matter^{op} some Matter^C
In Domain Of	has constituent^{op} has matter^{op}
In Range Of	has physical object^{op}

Resource^C

IRI	http://w3id.org/CE0N/ontology/resource0DP/Resource
Description	A resource able to be handled in the context of a circular value network, e.g. data generated by software systems in the CVN, materials or prooducts as physical objects handled in the CVN.
In Domain Of	hasResourceLocation^{op}
Super Class Of	Batch of objects^C Digital object^C Information^C Physical object^C Set of objects^C

Set of objects^C

IRI	http://w3id.org/CE0N/ontology/resource0DP/SetOfObjects
Description	A set of objects is a set of physical objects (items) that can be of different types, i.e. different kinds of items.
Sub Class Of	Resource^C has batch^{op} only Batch of objects^C <i>and</i> has batch^{op} some Batch of objects^C has physical object^{op} some Physical object^C <i>and</i> has physical object^{op} only Physical object^C

Geometry^c

IRI	http://www.opengis.net/ont/geosparql#Geometry
Description	A coherent set of direct positions in space. The positions are held within a Spatial Reference System (SRS).
In Domain Of	as GML ^{dp} as GeoJSON ^{dp} as WKT ^{dp}
In Range Of	hasResourceLocation ^{op}

Object Properties

contains information^{op}

IRI	http://w3id.org/CE0N/ontology/resourceODP/containsInformation
Sub Property Of	hasPart ^{op}
Domain	Information ^c
Range	Information ^c

has batch^{op}

IRI	http://w3id.org/CE0N/ontology/resourceODP/hasBatch
Description	hasBatch intends to represent that a set or batch of objects can be captured by a number of batches where each batch contains a number of physical objects.
Domain	Batch of objects ^c or Set of objects ^c
Range	Batch of objects ^c

has constituent^{op}

IRI	http://w3id.org/CE0N/ontology/resourceODP/hasConstituent
Description	hasConstituent intends to represent that a physical object can have a collection of composing components.
Sub Property Of	hasPart ^{op}
Domain	Physical object ^c
Range	Constituent ^c

has matter^{op}

IRI	http://w3id.org/CE0N/ontology/resource0DP/hasMatter
Description	hasMatter intends to represent that a physical object can have a collection of matter.
Domain	Physical object ^c
Range	Matter ^c

has part^{op}

IRI	http://w3id.org/CE0N/ontology/resource0DP/hasPart
Super Property Of	• containsInformation^{op} • has constituent^{op}

has physical object^{op}

IRI	http://w3id.org/CE0N/ontology/resource0DP/hasPhysicalObject
Description	hasPhysicalObject intends to represent that a batch of objects or a set of objects can have composing components of physical objects.
Domain	Set of objects ^c or Batch of objects ^c
Range	Physical object ^c

has resource location^{op}

IRI	http://w3id.org/CE0N/ontology/resource0DP/hasResourceLocation
Domain	Resource ^c
Range	Geometry ^c

is about^{op}

IRI	http://w3id.org/CE0N/ontology/resource0DP/isAbout
Domain	Information ^c

is realization of^{op}

IRI	http://w3id.org/CE0N/ontology/resource0DP/isRealizationOf
Range	Information ^c

Datatype Properties

batch size^{dp}

IRI	http://w3id.org/CE0N/ontology/resource0DP/batchSize
Description	batchSize intends to represent how many physical objects belong to a batch of objects.
Domain	Batch of objects ^c
Range	xsd:nonNegativeInteger

as GML^{dp}

IRI	http://www.opengis.net/ont/geosparql#asGML
Is Defined By	http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal
Description	The GML serialization of a Geometry.
Domain	Geometry ^c
Range	geo:gmlLiteral

as GeoJSON^{dp}

IRI	http://www.opengis.net/ont/geosparql#asGeoJSON
Is Defined By	http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal
Description	The GeoJSON serialization of a Geometry.
Domain	Geometry ^c
Range	geo:geoJSONLiteral

as WKT^{dp}

IRI	http://www.opengis.net/ont/geosparql#asWKT
Is Defined By	http://www.opengis.net/spec/geosparql/1.0/req/geometry-extension/geometry-as-wkt-literal
Description	The WKT serialization of a Geometry.
Domain	Geometry ^c
Range	geo:wktLiteral

Annotation Properties

description^{ap}

IRI <http://purl.org/dc/elements/1.1/description>

contributor^{ap}

IRI <http://purl.org/dc/terms/contributor>

created^{ap}

IRI <http://purl.org/dc/terms/created>

creator^{ap}

IRI <http://purl.org/dc/terms/creator>

description^{ap}

IRI <http://purl.org/dc/terms/description>

license^{ap}

IRI <http://purl.org/dc/terms/license>

title^{ap}

IRI <http://purl.org/dc/terms/title>

preferred namespace prefix^{ap}

IRI <http://purl.org/vocab/vann/preferredNamespacePrefix>

preferred namespace uri^{ap}

IRI <http://purl.org/vocab/vann/preferredNamespaceUri>

covers requirements^{ap}

IRI <http://www.ontologydesignpatterns.org/schemas/cpannotation/schema.owl#coversRequirements>

definition^{ap}

IRI <http://www.w3.org/2004/02/skos/core#definition>

pref label^{ap}

IRI <http://www.w3.org/2004/02/skos/core#prefLabel>

Namespaces

:

<http://w3id.org/CEON/ontology/resourceODP/>

dc <http://purl.org/dc/elements/1.1/>

dcterms <http://purl.org/dc/terms/>

geo <http://www.opengis.net/ont/geosparql#>

odp <http://www.ontologydesignpatterns.org/schemas/cpannotationschema.owl#>

owl <http://www.w3.org/2002/07/owl#>

prov <http://www.w3.org/ns/prov#>

rdf <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

rdfs <http://www.w3.org/2000/01/rdf-schema#>

skos <http://www.w3.org/2004/02/skos/core#>

vann <http://purl.org/vocab/vann/>

xsd <http://www.w3.org/2001/XMLSchema#>

Legend

c	Classes
op	Object Properties
dp	Datatype Properties
ap	Annotation Properties